



INTRODUCTION TO GAMES

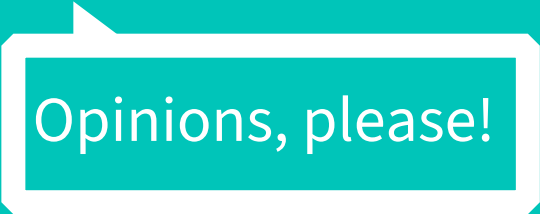
Part 7: Game Interfaces

Today's Lecture: Interface Design

1. What's an interface?
2. How do interfaces work?
3. Types of interfaces.
4. Creating interfaces.
5. Interface analyses.

1.

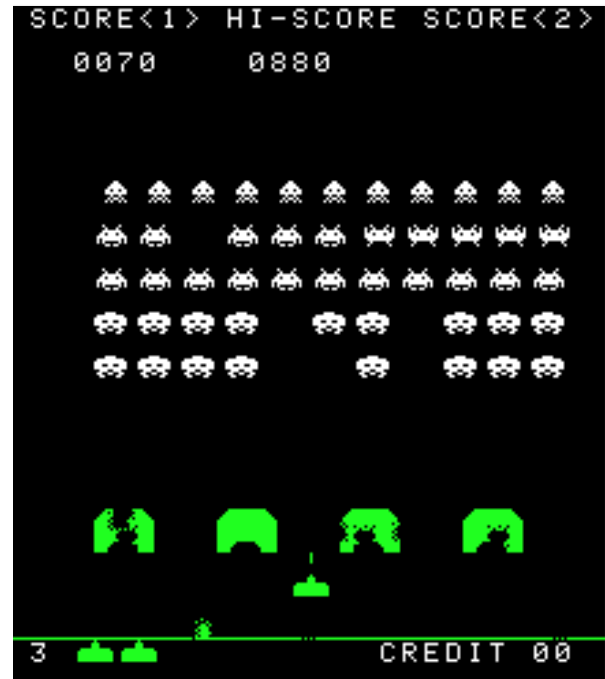
What's an Interface?



Opinions, please!

Is There an Interface?

Does this Space Invaders screenshot exhibit an **interface**?





Item Crafting

All Items



Neutralizing Boluses



Select item to craft

(A) :OK (B) :Back (X) :Switch display (mid) (Y) :Switch view (right) (LT) (RT) :Scroll (RT) :Simple view (C) :Help

Neutralizing Boluses

Alleviates poison buildup and cures poison

Consumable

No. Held 9 / 99 Stored

Required Items



Herba
Found in thickets and elsewhere

No. Required (No. Held)

1 (6)



Cave Moss
Gives off pale light in dark caves

1



Great Dragonfly Head
Found by hunting large dragonflies

1 (1)

Character Status

Level	73
Runes Held	630
Vigor	33
Endurance	10
Strength	20
Dexterity	34
Intelligence	30
Faith	9

HP	886 / 1132
FP	53 / 68
Stamina	108

Equip Load	49.2 / 76.2
Med. Load	

Poise	40
Discovery	107.0
Memory Slots	3



Elden Ring
Crafting menu

World Completion

99%

300/303 189/189
486/486 226/226
653/653

Queensdale

Completion

100%

17/17 7/7
16/16 9/9
21/21

Western Divinity Dam

Shaemoor Fields

Guild Wars 2
Map with important
locations

Sha
Ga



World of Warcraft (Old) User interface

[51:01] <Lady Deathwhisper> Casting Frostbolt: 2.0 sec
[51:01] <Lady Deathwhisper> Touch of Insignificance on Xarcy (1)
[51:02] Lady Deathwhisper yells: All part of the Master's plan....
Your end is inevitable....
[51:02] <Deadly Boss Mods> Lady Deathwhisper down after 10
minutes and 3 seconds!
[51:02] You receive Hum: (Emblem of Frost) x2.
[51:02] Reputation with The Ashen Verdict increased by 250.
[51:02] [R] [80:Ranzar]: *** Lady Deathwhisper has been defeated
see

Minlak 100%	Braknight 100%	Torr 91%	rip-Zeroaster 0%	Hammerstein 79%
Xarcy 100%	Slyminx 100%	Schalx 75%	Rosabella 100%	Madam 80%
+11.9k Ranza 122%	Nitechill 100%	ataclizmic 100%	Brenbenn 100%	Xandia 100%
Suigeneriss 99%	Whodareswins 100%	Spools 78%	Melstock 84%	rip-Akahotz 0%
Morark 100%	rip-Moods 0%	Savarak 100%	Allarganis 99%	lingledous 98%

Main Tanks

Ranza
122%

Xarcy
100%

Minlak
100%

Private Tanks

Toe-Brenbenn
100%

Abominable Might
Borrowed Time

Shac...

Chill of the Throne
Exhaustion

Weakened Soul

23:49:02 [6] [80:Madam5]: they don't say " sorry we don't have any
big mac's when I ask for one"
[23:49:03] [6] [80:Hammerstein5]: lol
[23:49:09] [6] [80:Xandins5]: you should wait till they give you one and
then say you wanted a big now
[23:49:17] [6] [80:Xandins5]: one
[23:49:31] [7] [Morark]: talk channel.
[23:49:45] [6] [80:Hammerstein5]: makes you wanna point to the picture
behind them and discuss why the burger they gave you is so
different.
[23:49:53] [7] [Morark]: joined channel.

A screenshot from the video game Dead Space. The player character, Isaac Clarke, is seen from behind, wearing his iconic dark, heavily detailed space suit with glowing green lights on the back. He is in a dark, industrial, and bloody environment. In the background, there are large red hoses hanging from the ceiling, a small white robot-like figure, and a large, bloody, dismembered body lying on a metal platform. A speech bubble with a white border is overlaid on the right side of the image, containing the text "Dead Space" in red and "Where's the interface?" in white.

Dead Space

Where's the
interface?

Is This an Interface Too?

What about this
PlayStation 5 DualSense
controller? Is it an
interface?



Skeleton



Xbox One Kinect

A different kind
of onput



INTERFACE

A definition attempt.



*The purpose of an interface is to allow players to communicate with the software (i.e., the game). Interfaces thus have two primary functions: **feedback** (providing information to the player; output) and **control** (receiving information from the player; input). All elements of a game interface further at least one of these two functions.*

Feedback: Status and Progress

Strategies

What should I **do** next?

Duration

How much longer until I achieve my goal?

Success

Am I **winning**?



Feedback: Teaching Concepts

Teach the **rules** of the game.

Include **instructions** in the interface.



Control

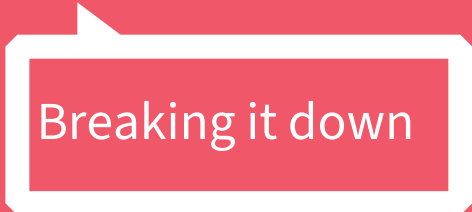
Mostly **physical interfaces** (controllers, etc.), but also virtual ones:

- buttons,
- touch input,
- text input,
- prompts.



2.

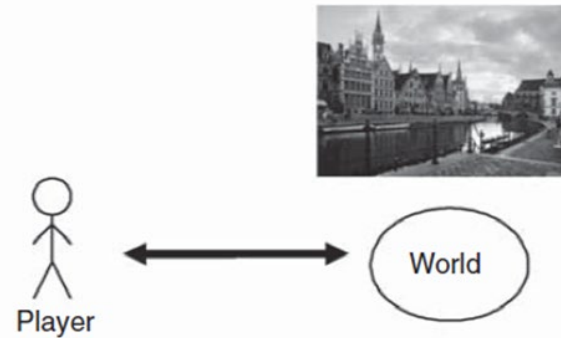
How Do Interfaces Work?



Breaking it down

Simply Put...

An interface is **everything between** the player and the game world.



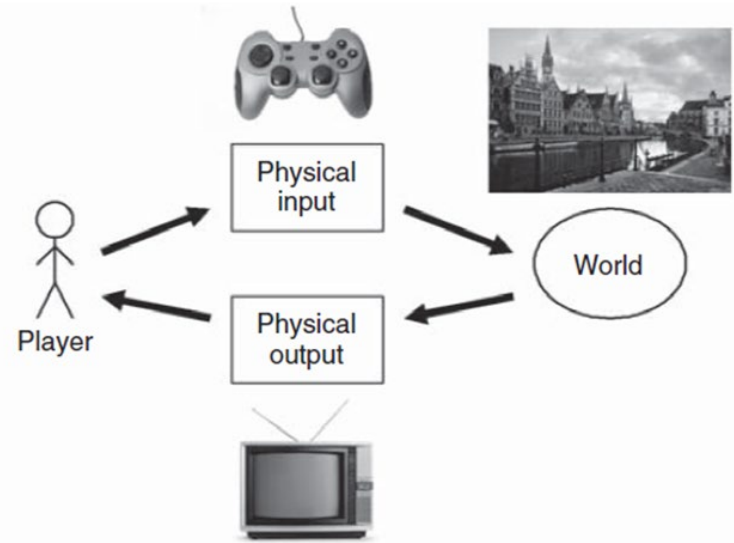
On A Closer Look...

Physical Input

The player manipulates the game world's status.

Physical Output

The player can perceive the results of their actions in a game.



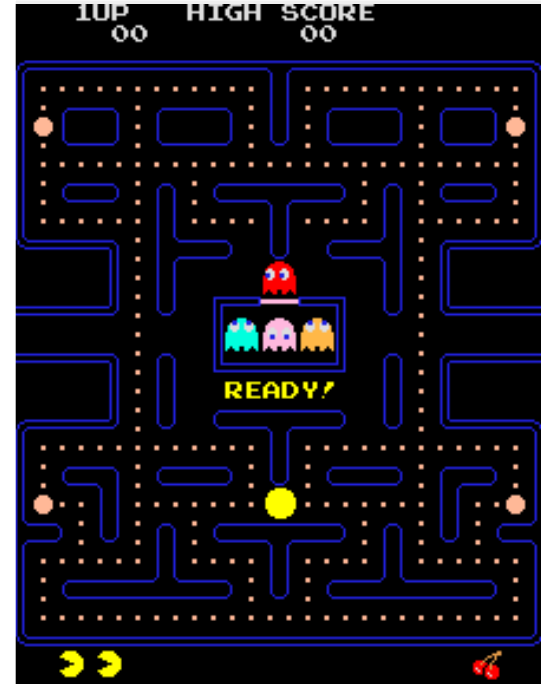
But There Is Something Missing

There's not always a direct connection between the player and the game world.

Virtual interface

An **intermediate** layer.

E.g., score display or menus.



The Virtual Interface

Input elements

E.g., a virtual menu.

Output elements

E.g., a score display.

It can be thin or dense.



The Complete Picture: Mapping

Data transformation

6 distinct directions:

Physical Input → World

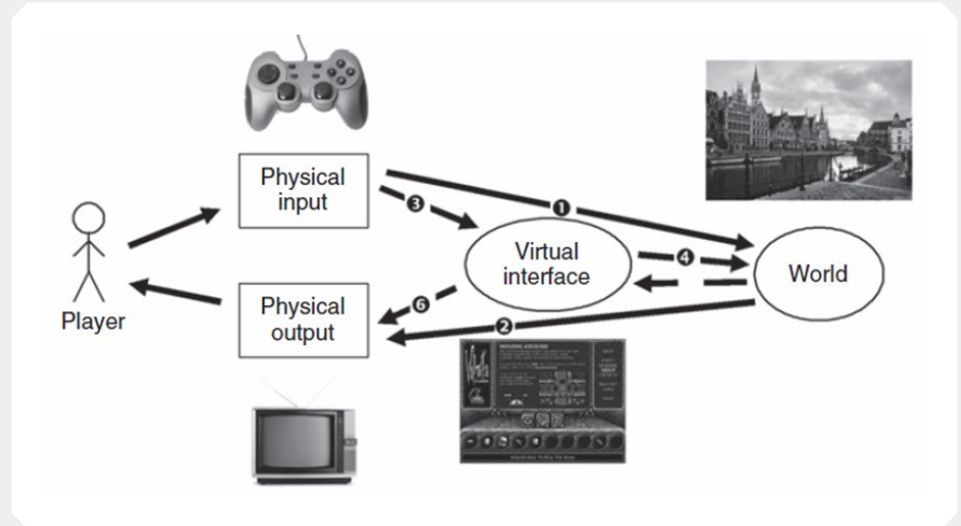
World → Physical Output

Physical Input → Virtual Interface

Virtual Interface → World

World → Virtual Interface

Virtual Interface → Physical Output



Physical Input → World

Pressing the analog stick moves the character.

The mapping decides **how fast it goes** or **how quickly it stops**.



World → Physical Output

Many games do not show the whole game world at once.

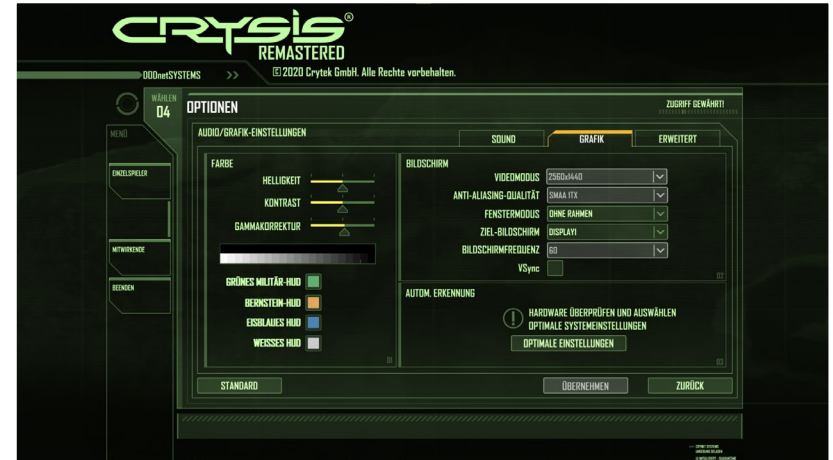
The mapping defines **which parts are visible** on the display.



Physical Input → Virtual Interface

How do controls influence the virtual interface?

The mapping defines, e.g., what **clicking**, **double-clicking**, etc., does.



Virtual Interface → World

How is the game world influenced when the player manipulates the virtual interface?

The mapping defines **when** actions occur or **how significant** their effects are.



World → Virtual Interface

How are changes in the world shown in the virtual interface?

The mapping defines **when** messages, effects, mode changes, and other events are displayed.



Virtual Interface → Physical Output

What data is shown to the player in what way?

The mapping defines the position on the screen, font size, colors, and other settings.



The Ultimate Goal: A Transparent Interface

Does the interface let players **do what they desire**?

Is it **simple enough** that players will use it without thinking (with enough practice)?

Do new players find it **intuitive**?



3.

Types of Interfaces

How do virtual interfaces manifest in games?

Concept: Diegesis

		Is the representation visualized in the 3D game space?	
		no	yes
Is the representation existing in the fictional game world?	no	non-diegetic representations	spatial representations
	yes	meta representations	diegetic representations

Diegetic Interfaces

The interface is **part of the game world**.

Avatars and NPCs can perceive it.

		Is the representation visualized in the 3D game space?	
		no	yes
Is the representation existing in the fictional game world?	no	non-diegetic representations	spatial representations
	yes	meta representations	diegetic representations



Non-diegetic Interfaces

The interface is **outside of the game world**.

Only visible and audible to players in the real world.

Classic heads-up displays.

		Is the representation visualized in the 3D game space?	
		no	yes
Is the representation existing in the fictional game world?	no	non-diegetic representations	spatial representations
	yes	meta representations	diegetic representations



Spatial Interfaces

The **UI element** is
represented in 3D space.

It is usually not part of the
game world.

		Is the representation visualized in the 3D game space?	
		no	yes
Is the representation existing in the fictional game world?	no	non-diegetic representations	spatial representations
	yes	meta representations	diegetic representations



Meta Interfaces

Exist in the game world, but are usually not represented in the game space.

Blood spatter, computer screens, etc.

		Is the representation visualized in the 3D game space?	
		no	yes
Is the representation existing in the fictional game world?	no	non-diegetic representations	spatial representations
	yes	meta representations	diegetic representations



4.

Creating Interfaces

The process of making it happen

Roles involved in the Process

- **Designers** (interface designer, game designer)
 - How will the interface work?
- **Developers**
 - How will the interface be incorporated into the game?
- **Artists** (graphic, sound)
 - How will the interface look and sound?
- **Testers/Quality Assurance** and **Producers**
 - Ensure that the interface will work as expected.

Approach 1: “Steal”

Top-down approach

- Get **inspired** by successful games in a similar genre.
- Don't copy their interfaces exactly!
- Little changes can have a significant impact.

[GAMES](#)[SCREENSHOTS](#)[ARTICLES](#)[ABOUT](#)[WIREFRAME MOCKUP](#)

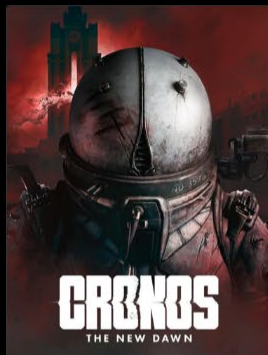
Recent games add



Dying Light: The B...

SEPTEMBER 18, 2025

ACTION



Cronos: The New ...

SEPTEMBER 5, 2025

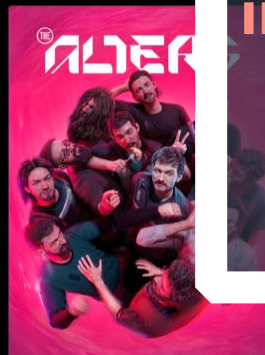
ACTION



Witchfire

SEPTEMBER 20, 2023

FPS



The Alters

JUNE 12, 2025

INDIE

ACTION

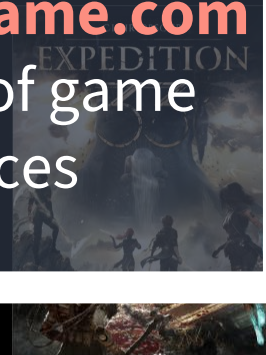
interfaceingame.com
Collection of game
interfaces

Against the Storm

OCTOBER 18, 2021

STRATEGY

SIMULATION



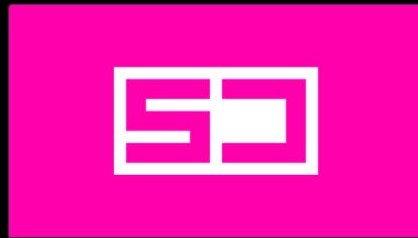
Clair Obscur: Exp...

APRIL 24, 2025

RPG

ACTION

Recent articles

[ALL ARTICLES](#) →

Screen Categories

- All Gameplay & HUDs
- Maps
- Items & Unlocks
- Tutorials
- Buttons & Controls

Title & System Menus

TITLE

- Title Screen
- Mode & Screen Select
- Game Difficulty
- Load/Save
- Loading Screen
- Credits

SETTINGS

- Settings Menu
- Settings Options
- Button Layouts

MODALS AND OVERLAYS

- Cutscenes & Story
- Dialogue & Speech
- Guided Tutorial
- Modals & Popups
- Character Intro
- Progress/Feature Unlocked

Game States

Stats and Resources

Information

Meta-Game Features

RECENTLY ADDED



UNDER NIGHT IN-BIRTH II Sys:Celes
12 hours ago



Dispatch
Yesterday



Sonic Racing: CrossWorlds
2 days ago

gameuidatabase.com
Collection of game
interfaces



Planet Coaster 2
4 days ago



Yooka-Laylee
5 days ago

NEW & FEATURED



Hollow Knight: Silksong
Released Sep 2025



Ghost of Yōtei
Released Oct 2025



Hades II
Released Sep 2025



Absolum
Released Oct 2025



Marvel Tokon: Fighting Souls (Closed Beta)
Released Oct 2025



Sonic Racing: CrossWorlds
Released Sep 2025



Cronos: The New Dawn
Released Sep 2025



PowerWash Simulator 2
Released Oct 2025



Yooka-Replaylee
Released Oct 2025



Dying Light: The Beast
Released Sep 2025



Silent Hill f
Released Sep 2025



Approach 2: Customize

Bottom-up approach

- Design your interface **from scratch**.
- Try to come up with new solutions for old problems.
- It's time-consuming and needs experience!

Interface Design Documentation

Created by the designers

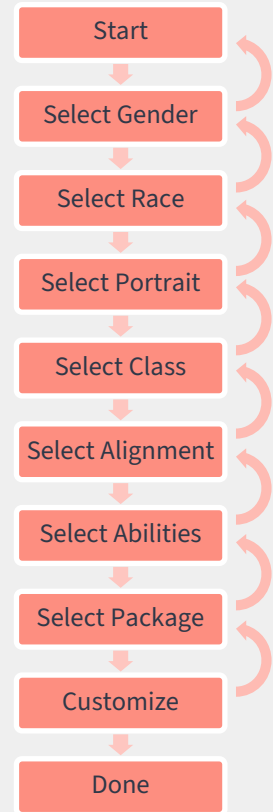
- Defines the way the interface works in its entirety.
- It's the foundation for developers and artists.
- Different tools are available.

Flow Diagram

Explains the sequential nature of an interface.

- Which screens does it contain?
- In what order will they appear?

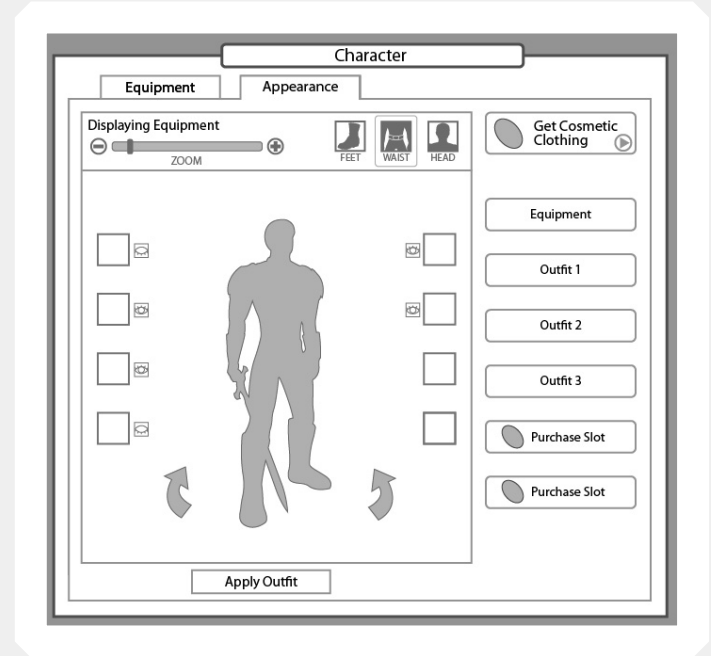
Great for interactive menus!



Sketches & Wireframes

Illustrate the interface's
layout and proportions.

Focus on the usability
perspective, not aesthetics.
They are used primarily for
non-diegetic and meta
interfaces.

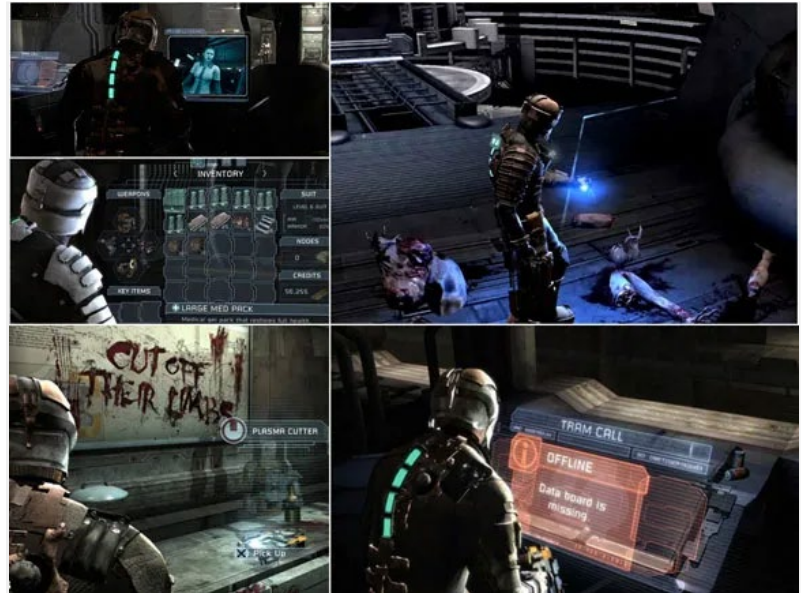


(Fake) Screenshots

Illustrate how diegetic, spatial, and meta interfaces can look.

Use your game's art.

If not ready, use similar art from other games.



Technical Specifications

Explain events that the interface will trigger.

- Supplementary to flow charts and wireframes.
- “Mouse over button highlights the button and plays a sound.”
- “Button A triggers a jump of the character (animation and sound).”

Interface Production Pipeline

Creating a game's interface is an iterative process.

- Usually, several iterations are necessary.
- Problem: It's one of the first components necessary, but one of the last that can be evaluated appropriately (when the game is finished).

Interface Pipeline: Steps Involved

1. Create the interface design documentation.
2. Review (developers, artists) – is it feasible?
3. Artists create graphical mockups; programmers create a prototype.
4. Review (team, testers/QA) – do the proportions of the layout work?
5. Developers implement it, and artists create the final artwork.

Interface Pipeline: Steps Involved

6. Extensive testing (e.g., focus groups with external people).
7. Incorporate the feedback and generate the final interface.

Considerations for Interfaces



Functionality

Does the interface accomplish what it's supposed to do?



Usability

Is the interface usable?



Accessibility

Does the interface consider people with disabilities?



Immersion

Is the interface transparent?



Aesthetics

Does the interface match the game's visual style?

Functionality

Does the interface provide the necessary **feedback and control**?

Is the current **status** clearly visible?

Is the interface **reactive**?

Are the player's **goals** (short-term, mid-term, long-term) identifiable?



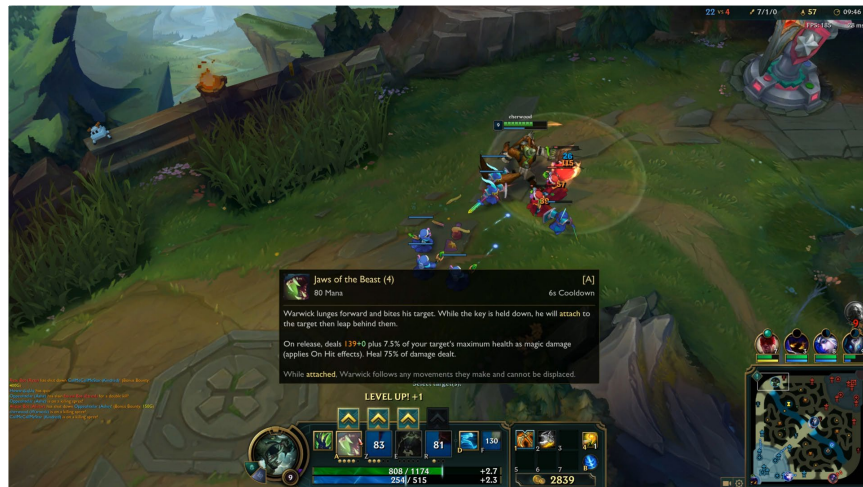
Usability

Is the interface **user-friendly**?

Are important actions **easy to invoke**?

Does it provide enough **feedback** (progress bars, tooltips, etc.)?

Is it **consistent**?



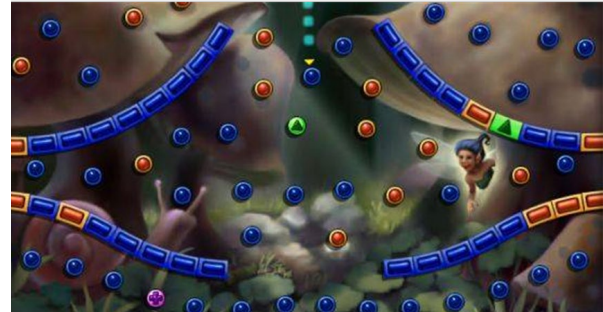
Accessibility

Think of players with
special needs!

Vision (larger text size,
colorblind modes, shapes).

Hearing (subtitles for
dialogs).

Physical (allow for a wide
range of controllers).



Immersion

An interface should not distract.

Aim for a **transparent** interface.

Use diegetic and meta interfaces.



Aesthetics

The interface should support the game's mood and atmosphere.

Match colors and typography.

Work with icons to reduce clutter.



There's Still More!

Audio

- Can be used to support an interface (reinforce the visuals).
- Can also be a separate type of interface (e.g., accessible games).

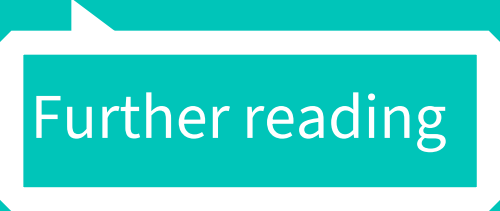
There's Still More!

“New” technologies

- Touchscreens: A hybrid form of control and feedback.
- Gesture controls: Motion sensing, facial recognition, eye-tracking.
- Speech recognition: Voice-controlled input.
- Electroencephalography, galvanic skin response: control through the mind, feedback from the body.

5.

Interface Analyses



Further reading

Interface Analysis Example

Game UI Discoveries: What Players Want

Analyzes interfaces of Far Cry 2, Dead Space, Team Fortress 2, and World of Warcraft

<https://www.gamedeveloper.com/design/game-ui-discoveries-what-players-want>



Interface Analysis Example

Mass Effect: Massive Interface Fail

Deals with character customization, equipment, shop menu, galaxy map, and journal

<http://gamedesignreviews.com/reviews/mass-effect-interface-fail/>



Interface Analysis Example

9 ways The Last of Us Part II and Ghost of Tsushima increase player immersion through their User Interface (UI)

How these two games get it right

<https://andyscabin.games.wordpress.com/2021/05/24/how-the-the-last-of-us-part-ii-and-ghost-of-tsushima-increase-player-immersion-through-their-user-interface-ui/>



Interface Analysis Example

A UX Analysis of Cyberpunk 2077's HUD

UI analysis and suggestions
for improvement

<https://medium.com/super-jump/a-ux-analysis-of-cyberpunk-2077s-hud-f74afe6b9961>

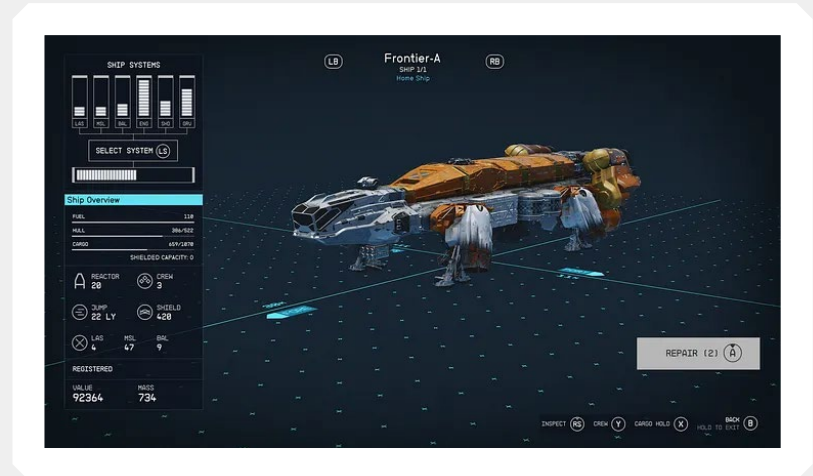


Interface Analysis Example

Lost in Space: The UX Problems Plaguing Starfield

Issue analysis of Starfield's inventory and menus

<https://medium.com/@ryanquintal/lost-in-space-the-ux-problems-plaguing-starfield-efc8ec9b9797>





“

A user interface is like a joke. If you have to explain it, it's not that good.

– Martin LeBlanc

<https://x.com/martinleblanc/status/466638260195041280>

Credits

- S. 4: Space Invaders: <https://en.wikipedia.org/wiki/File:SpaceInvaders-Gameplay.gif>
- S. 5: Elden Ring: <https://www.windowcentral.com/elden-ring-how-craft-items>
- S. 6: Guild Wars 2: <https://guildjen.com/new-players-guide-leveling-to-80-and-getting-your-gear/>
- S. 7: World of Warcraft: <https://media.indiedb.com/images/articles/1/118/117437/auto/oldui.jpg>
- S. 8: Dead Space: https://www.pcgamehunters.de/galerieupload/galerie_Dead%20Space_1225640893.jpg
- S. 9: DualSense Controller: https://de.wikipedia.org/wiki/DualSense#/media/Datei:Playstation_DualSense_Controller.png
- S. 10: Xbox One Kinect: <https://www.spieletipps.de/bilderstrecke/2423/1/>
- S. 13: Soul Calibur IV: <https://www.fightersgeneration.com/np5/gm/sc4-s6.jpg>
- S. 14: Super Mario World: <https://www.nintendo.de/Spiele/Super-Nintendo/Super-Mario-World-752133.html#Galerie>
- S. 15: Call of Duty Advanced Warfare: <https://knowyourmeme.com/photos/1488684-press-f-to-pay-respects>
- S. 19: Pacman: <https://en.wikipedia.org/wiki/Pac-Man#/media/File:Pac-man.png>
- S. 20 Gears of War 3: <https://gameaccessibilityguidelines.com/gears-of-war-3-shot-indicator/>
- S. 22: Super Mario Galaxy: <https://www.videogameszone.de/Super-Mario-Galaxy-Spiel-35946/News/Wii-Bilder-in-HD-Super-Mario-Galaxy-Screenshots-in-720p-689134/galerie/1155254/>
- S. 23: Fallout New Vegas: https://www.gamestar.de/galerien/fallout_new_vegas,91491.html

Credits

- S. 24: Crysis Remastered: <https://www.gamingmithandicap.de/die-barrierefreiheit-einstellungen-von-crysis-remastered/>
- S. 25: Fallout 76: <https://www.rockpapershotgun.com/fallout-76-perk-cards-all-the-perks-how-to-upgrade-them>
- S. 26: Guacamelee: <https://www.gamespot.com/reviews/guacamelee-review/1900-6406634/>
- S. 27: World of Warcraft: https://www.reddit.com/r/wow/comments/14blxe/i_was_sifting_through_some_old_screenshots_from/
- S. 28: Starfire Saga V: <https://www.penny-arcade.com/comic/2007/06/08/from-the-makers-of-beam-kings>
- S. 32: Team Fortress 2: https://wiki.teamfortress.com/wiki/Heads-up_display/de#/media/File:CTFHud.png
- S. 33: World of Warcraft: https://cuppyville.files.wordpress.com/2013/09/wowscrnshot_092913_150141.jpg
- S. 34: Call of Duty: Modern Warfare 2: <https://aspiringgameprogrammer.files.wordpress.com/2009/11/vlcsnap-112634.png>
- S. 42: Wireframe Example: http://itfe.co.kr/education_/img/img02_13_5_3.jpg
- S. 43: Dead Space Collage: <https://www.gamedeveloper.com/design/game-ui-discoveries-what-players-want>
- S. 49: Portal: <https://www.videogameszone.de/Portal-Still-Alive-Spiel-28237/Tests/Portal-Still-Alive-im-Arcade-Test-669740/galerie/948571/>
- S. 50: League of Legends: <https://interfaceingame.com/screenshots/league-of-legends-spell-hover/>
- S. 52: Neverwinter Nights 2: <https://lparchive.org/Neverwinter-Nights-2/Update%2031/>
- S. 53: Joe Danger: <https://www.gamerevolution.com/review/65902-joe-danger-special-edition-review>